

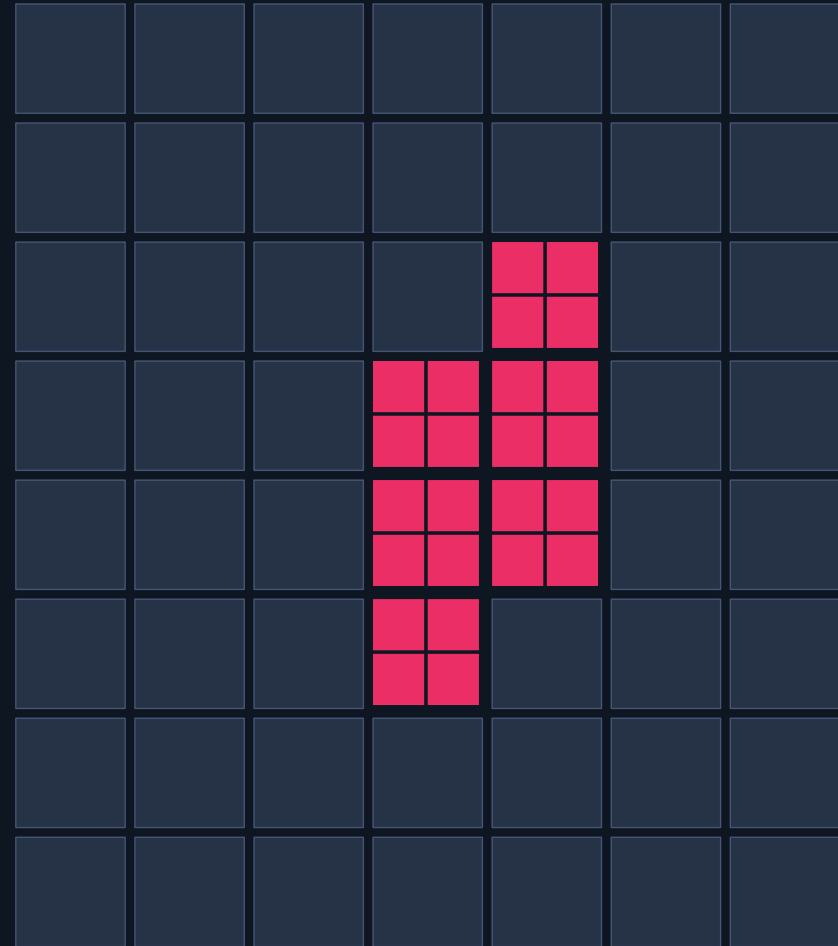
PROJECT PROPOSAL · 10 MINUTES · JULY 14

Perception-Aware Dynamic Diffusion Transformers

for Safety-Critical Scene Generation

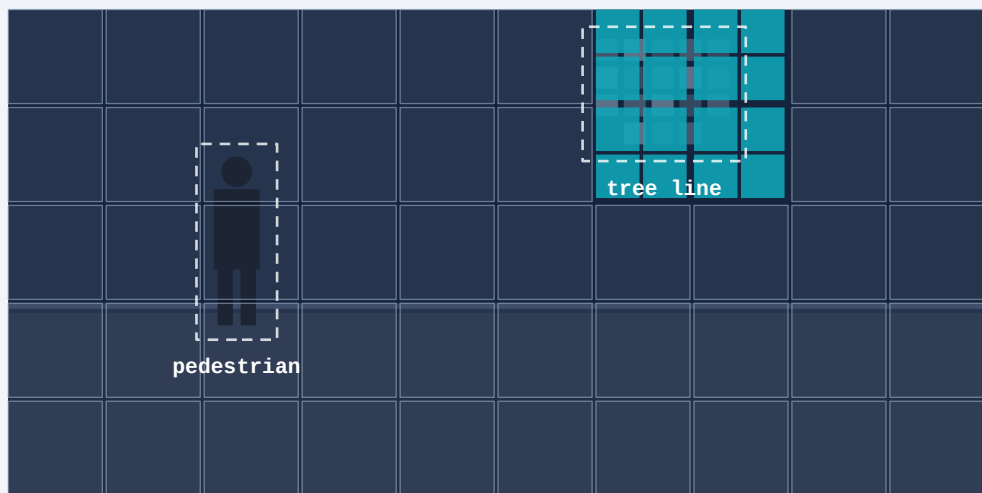
*Entropy tells the model where the pixels are busy.
It never tells it where the pedestrian is.*

[Your Name] · [Course / Advisor]



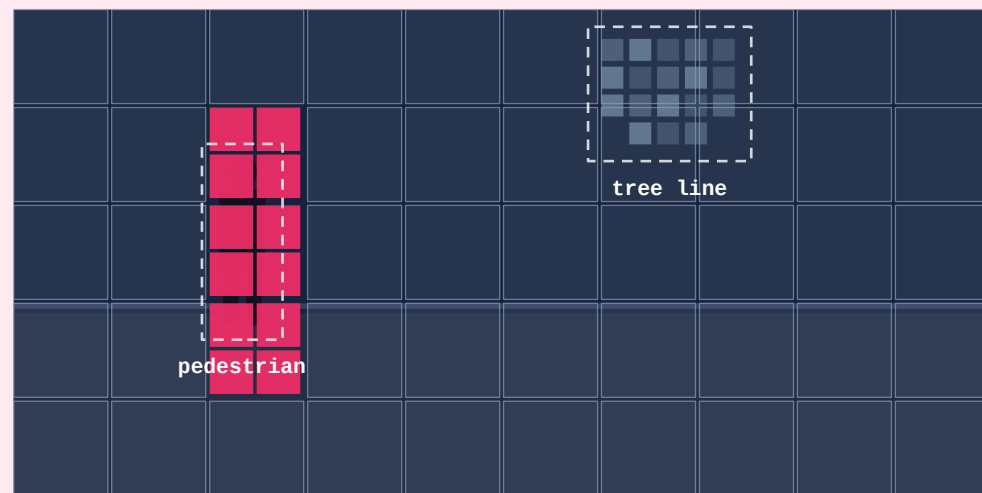
The problem: entropy is not importance

WHAT ENTROPY SEES



Fine capacity goes to the textured tree line.
The dark-clothed pedestrian stays coarse.

WHAT ACTUALLY MATTERS

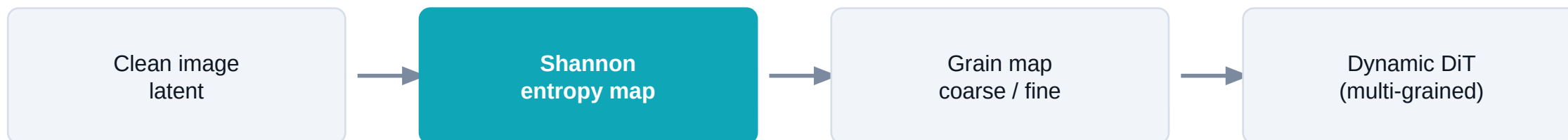


Fine capacity goes to the vulnerable road user.
The tree line can stay coarse — nobody brakes for it.

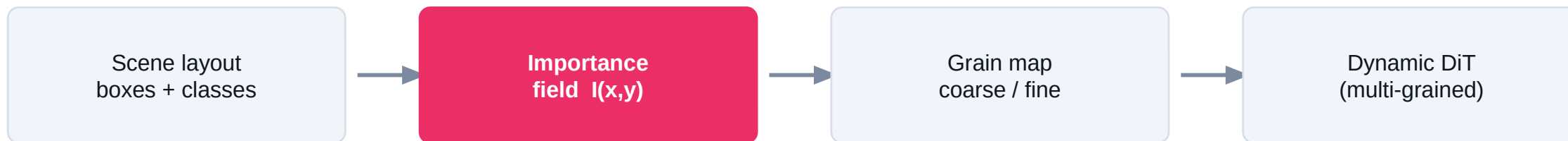
Shannon entropy is a texture detector. Safety is a semantic property.

What the project is: swap the signal, keep the machinery

D²iT (BASELINE)



PA-D²iT (OURS)



One block changes. Everything downstream — the dynamic transformer, the multi-grained loss — is inherited from D²iT. That is what makes the hypothesis cleanly testable at matched compute.

Problem statement: generative capacity is allocated by a visual proxy, not by what the scene actually risks.

Previous related work — and the gap



D²iT (2025)

Dynamic Diffusion Transformer.
Hierarchical VAE + entropy-driven
coarse/fine grain map.

*Direct predecessor — we replace its
signal.*



Layout — driving scenes

GeoDiffusion, MagicDrive,
DrivingDiffusion, Panacea,
BEVGen.

*Layout conditioning exists. It steers
content, not capacity.*



Layout — image (general)

LayoutDiffusion, GLIGEN,
ControlNet.

*Grounding a diffusion model on
boxes is solved.*



Adaptive compute

Token pruning/merging, region-
adaptive sampling, diffusion-MoE
routers.

*Saliency is used to SAVE compute,
not to spend it where risk is.*

THE GAP

Nobody uses semantic importance to drive VARIABLE GENERATIVE CAPACITY, and nobody evaluates whether the safety-critical objects actually came out better. That is the contribution — the integration, not the ingredients.

Why this matters, and who it helps

1

Why this problem

Rare, low-contrast, high-stakes cases — a pedestrian at night, a cyclist in rain — are exactly where real driving data is thinnest and where generative models are weakest. Entropy-driven allocation makes it worse, not better.

2

The impact

Better synthetic training data where detectors actually fail. Object-region fidelity translates directly into downstream detector mAP — the metric that matters, not a prettier global FID.

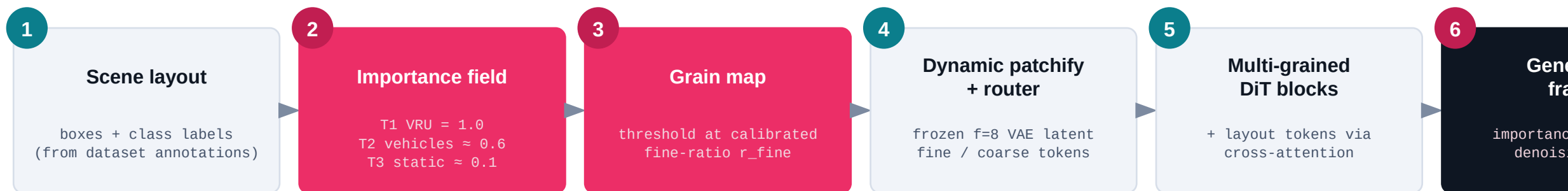
3

Who benefits

AV perception teams (targeted data augmentation for long-tail cases) · safety validation and scenario testing · any researcher who needs generative capacity spent by semantics rather than texture.

A few hundred pixels decide whether the car brakes. Those are the pixels the model should be spending its capacity on.

Method: the PA-D²iT pipeline



Key property: the grain map is derived from the LAYOUT, not from the noisy latent. It is therefore deterministic and identical at every timestep — which sidesteps the noisy-routing failure that plagues diffusion-MoE routers at high t .

V-Lite · RECOMMENDED

Freeze a pretrained f=8 SD/LDM VAE. Add dynamic patchify + importance router on the latent. All effort goes into the diffusion-side hypothesis. Fits a Colab Pro budget.

V-Full · STRETCH

Retrain the hierarchical D-VAE with $F = \{8, 16\}$ as in D²iT. A truer test of the multi-grain code — but it spends most of the compute budget on VAE pretraining.

Loss: importance-weighted multi-grained denoising — D²iT's per-grain α , scaled by a safety weight $w \in \{1, 2, 5\}$ on VRU regions.

Datasets and models

DATASETS

● Cityscapes

Primary. Clean instance boxes, manageable scale, standard baselines. Everything is developed here.

● BDD100K — night subset

The stress test. Exactly the low-entropy / high-importance regime the method is designed for.

● nuScenes

Stretch. 3D boxes → richer layouts, multi-view. Only if the first two land.

MODELS CONSIDERED

● Backbone

DiT-B class transformer, $\approx 136\text{M}$ params, latent diffusion on a frozen $f=8$ VAE. DDPM training, DDIM sampling.

● Ours: PA-D²iT

V-Lite — dynamic patchify + layout-derived importance router + layout cross-attention + weighted loss.

● Compute

Colab Pro. bf16, gradient accumulation, gradient checkpointing, EMA, resume-from-checkpoint to Drive. Runs sized to fit 12–16 GB.

Every model is trained at MATCHED COMPUTE. This is a proof of concept about allocation — not a bid for state of the art.

Evaluation metrics and baseline systems

EVALUATION METRICS

Object-region FID ↓

PRIMARY

FID computed on cropped VRU regions, not the whole frame. Directly measures the thing we claim to improve.

Downstream detector mAP ↑

PRIMARY

Train a detector on generated data, test on real. The honest end-to-end test of usefulness.

Global FID ↓

GUARDRAIL

Must not regress. Proves we bought object fidelity without wrecking the background.

BASELINE SYSTEMS (all matched compute)

DiT-B

Vanilla. No layout, uniform capacity.

DiT-B + layout

Layout-conditioned, still uniform capacity. Isolates the value of conditioning alone.

D²iT-B

Entropy-driven grain. THE baseline to beat — same machinery, different signal.

PA-D²iT-B (ours)

Importance-driven grain.

H1 PA-D²iT beats D²iT on object-region FID and downstream mAP at equal compute and no worse global FID. If it doesn't, the entropy proxy was good enough — and that is a publishable negative result, not a failure.

What can go wrong — and the mitigation

RISK	MITIGATION
1 Importance loses to entropy <i>The honest one. Maybe texture is a good enough proxy.</i>	▶ Matched-compute design makes the comparison clean. Ablate the safety weight $w \in \{1,2,5\}$ and the fine-ratio r_{fine} . A well-ablated negative result still answers the question.
2 Global FID regresses <i>Backgrounds degrade as capacity is pulled toward objects.</i>	▶ Global FID is an explicit guardrail metric, not an afterthought. Tune r_{fine} to find the knee. Report the trade-off curve rather than hiding it.
3 Colab compute runs out <i>Session timeouts, GPU availability, long training runs.</i>	▶ Frozen VAE removes D-VAE pretraining entirely. bf16 + gradient accumulation + checkpointing to Drive; every run sized to fit 12–16 GB so a T4 still works.
4 Novelty overlap <i>Layout-conditioned driving generation already exists.</i>	▶ Position on capacity ALLOCATION, not conditioning — and say so on slide 4 before anyone asks. The DiT-B + layout baseline exists precisely to isolate this.
5 Scope creep <i>Three datasets, two VAE variants, a detector eval.</i>	▶ Go / no-go gates: Week 2 (pipeline + grain maps working) and Week 7 (does object-region FID move?). V-Full and nuScenes are explicitly stretch goals, cut first.

Timeline with go / no-go gates (~10 weeks)



The gates are the real risk control. Week 2: if grain maps aren't rendering from layouts, the method is dead and there is still time to pivot. Week 7: if object-region FID hasn't moved against D²iT-B, stop scaling and start ablating — the remaining weeks go into understanding WHY, which is the more valuable result anyway.

Cut order if time runs short: nuScenes → V-Full (hierarchical D-VAE) → BDD100K night → ship Cityscapes only.

IN ONE SENTENCE

Put the model's capacity where the risk is.

Replace D²iT's entropy signal with a layout-derived importance field, so that fine-grained generative capacity lands on pedestrians and cyclists instead of on textured sky — and then actually measure whether it did.

QUESTIONS

